

PHINMA UNIVERSITY OF ILOILO

College of Information Technology Education

CAPSTONE PROJECT PROPOSAL

WEB-BASED MR. SOFTY INTEGRATED BUSINESS INTELLIGENCE PLATFORM

In partial fulfillment for the requirements in Capstone 1 and
Research Subject

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CHAPTER 1

INTRODUCTION

In today's digital era, technology plays an important role in improving the efficiency and productivity of businesses. Many companies and establishments are now adopting computerized systems to make their operations faster, more accurate, and more organized. Through the use of technology, businesses can easily manage transactions, monitor inventory, track employee attendance, and generate reports that help in decision-making. Compared to manual methods, computerized systems provide better data management, reduce human errors, and improve the overall workflow of the business.

For small and medium-sized businesses, the use of information technology is becoming more important as business operations continue to grow. Manual processes may still be effective for small-scale operations, but they often become inefficient and difficult to manage when the volume of transactions increases. Traditional methods such as writing sales records in notebooks, manually checking inventory, and using logbooks for attendance tracking can consume a lot of time and may lead to inaccurate records, lost data, and delayed transactions. These problems can affect the performance and productivity of the business.

One of the common problems experienced by businesses that still rely on manual processes is inaccurate record management. Manual sales recording may result in incorrect computations and incomplete reports, while poor inventory monitoring can lead to overstocking or shortages of products. In addition, manual attendance tracking may cause errors in employee work records and difficulty in monitoring employee performance. Because of these challenges, businesses need a more efficient and reliable system that can automate their daily operations and provide accurate information in real time.

Mr. Softy, a local business, currently uses manual methods in managing sales transactions, employee attendance, and inventory monitoring. The business records sales manually, monitors inventory without an automated system, and tracks employee attendance using traditional methods. As a result, the business experiences difficulties in organizing records, retrieving information quickly, and maintaining accurate data. These challenges may slow down

business operations and increase the possibility of errors, especially as the business continues to expand.

To address these problems, this study proposes the development of a Web-Based Mr. Softy Integrated Business Intelligence Platform. The proposed system aims to automate important business processes such as sales recording, attendance tracking, inventory management, and report generation. The system will provide a centralized platform where business data can be stored, processed, and monitored efficiently. Through this platform, the business owner can access real-time information, minimize manual work, reduce errors, and improve decision-making.

The development of the proposed system is expected to help Mr. Softy improve the efficiency and accuracy of its daily operations. Furthermore, the study aims to demonstrate how web-based technologies and business intelligence solutions can help small businesses adapt to modern digital practices and achieve better operational management.

Statement of the Problem

Mr. Softy currently manages its sales, employee attendance, and inventory using manual methods. While this approach is simple, it creates several challenges that affect the efficiency and accuracy of the business operations. Manual processes are time-consuming, prone to errors, and make it difficult to access and organize data.

Another issue is that the business does not have a single system that connects all its operations. Because of this, information is scattered, and it becomes harder for the owner to monitor the business and make decisions based on reliable data.

With these concerns, this study aims to answer the following questions:

1. How can sales transactions be automated to reduce errors and improve the recording of daily sales?
2. How can employee attendance be monitored and recorded more accurately?
3. How can inventory be managed efficiently to avoid stock shortages and incorrect records?
4. How can integrating sales, attendance, and inventory into one system improve business operations?
5. How can the system help the business owner make better decisions?

Objectives of the Study

General Objectives

The main goal of this study is to develop a web-based system for Mr. Softy that will automate sales recording, employee attendance tracking, and inventory management to improve the overall efficiency and accuracy of the business.

Specific Objectives

Specifically, this study aims to:

1. To develop a system that can automatically record daily sales transactions to make the process faster and more accurate.
2. To create a feature that can track employee attendance in real time and reduce errors in recording work hours.
3. To design an inventory module that can monitor product stocks and help prevent shortages or overstocking.
4. To reduce errors and delays caused by manual processes in handling business data.
5. To provide organized and accessible records that can help the business owner make better decisions.

Scope and Delimitation

This study focuses on developing a web-based system for Mr. Softy that will manage sales, employee attendance, and inventory. The system will allow authorized users, such as staff and administrators, to record transactions, track attendance, and monitor product stocks.

The system will include features such as order recording, attendance logging, inventory tracking, and basic reporting. It will be accessible through a web browser, making it easy to use within the business.

However, the study has some limitations. It will only focus on internal business operations and will not include features such as online payments, delivery tracking, or mobile applications. Advanced financial reports and large-scale system deployment are also not included in this study.

Significance of the Study

The results of this study are expected to benefit the following:

Mr. Softy Business - The system will help the business manage sales records, employee attendance, and product inventory more efficiently.

Business Owner / Management - The system will provide organized and accurate data that can help the owner monitor sales, employee attendance, and product stocks.

Employees - The system will make attendance tracking easier and more accurate. It will help ensure that employees' work hours are properly recorded.

Customers - Although the system is mainly for internal use, improved business operations can lead to better service and faster transactions for customers.

Future Researcher - This study may serve as a reference for future researchers who plan to develop similar systems related to sales management, employee attendance monitoring, and inventory management.

Definition of Terms

Business Intelligence Platform - A system that analyzes business data to support smart business decisions.

Computerized Management System - A digital system used to manage and organize business operations.

Employee Attendance Tracking - The process of recording and monitoring employee work hours.

Inventory Management - The process of tracking and controlling product stock.

Manual System - A method of managing records without using automated computer systems.

Product Inventory - The products or goods available for sale in a business.

Sales Recording - The process of documenting sales transactions.

Stock Monitoring - The process of checking and updating product stock levels.

Web-Based System - A system that is accessed through a web browser using the internet or a network.

Authorized User - A person who has permission to access and use the system.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

Foreign Studies

Business Intelligence (BI) systems have become an important tool for organizations that aim to improve decision-making, operational efficiency, and data management. According to the study conducted by Joaquim Lapa, Jorge Bernardino, and Ana Figueiredo (2014), BI platforms are designed to collect, process, and analyze business data in order to generate meaningful information for organizations. Their study compared several open-source BI tools such as Actuate, JasperSoft, OpenI, Palo, Pentaho, SpagoBI, and Vanilla. The researchers emphasized that selecting the appropriate BI platform depends on the needs of the organization, including scalability, usability, and data integration capabilities. The study concluded that BI systems improve data accessibility, reporting accuracy, and operational efficiency, which are essential for business growth.

Lapa, J., Bernardino, J., & Figueiredo, A. (2014). Comparative analysis of open-source business intelligence platforms. *Proceedings of the International Symposium on Information Systems and Design of Communication*, 86-89.

<https://dl.acm.org/doi/abs/10.1145/2618168.2618182> ISBN 978-1-4503-2824-0

Wixom and Watson (2010) highlighted the importance of Business Intelligence in improving organizational performance. According to their research, BI systems help companies make data-driven decisions, improve productivity, and gain competitive advantage. The researchers explained that organizations using BI technologies are more capable of analyzing business operations and reducing dependence on manual processes, resulting in improved accuracy and efficiency.

Wixom, B. H., & Watson, H. J. (2010). The BI-based organization. *International Journal of Business Intelligence Research*, 1(1), 13-28. https://www.researchgate.net/publication/220673059_The_BI-based_organization

ISSN 1947-3591

Another related study by Chen, Chiang, and Storey (2012) discussed the evolution of Business Intelligence and analytics. The researchers explained that BI systems evolved from simple reporting tools into advanced analytics platforms capable of handling real-time data processing and predictive analysis. Their findings suggested that organizations adopting BI technologies are better prepared to respond to market changes and customer demands because of faster and more accurate access to business information.

Chen, H., Chiang, R. H. L., & Storey, V. C. (2012). Business intelligence and analytics: From big data to big impact. *MIS Quarterly*, 36(4), 1165-1188. <https://www.jstor.org/stable/41703503>
ISSN 0276-7783

Furthermore, the study conducted by Turban, Sharda, and Delen (2011) focused on the role of Business Intelligence systems in organizations and small businesses. The researchers stated that BI technologies improve decision-making by providing accurate reports, data visualization, and business analysis tools. The study also emphasized that integrating business operations into one centralized system improves coordination, communication, and operational efficiency within organizations.

Turban, E., Sharda, R., & Delen, D. (2011). *Decision support and business intelligence systems*. Pearson Education.
https://www.researchgate.net/publication/334485412_Open_Source_Business_Intelligence_on_a_SME_A_Case_Study_using_Pentaho
ISBN 978-0136107293

Local Studies

In the Philippine context, several studies have explored the use of computerized systems and business technologies among small and medium enterprises (SMEs). These studies emphasized the importance of automated systems in improving operational efficiency, accuracy, and business management.

A study entitled "*Inventory and Point-of-Sale System for Bida Mart*" discussed the challenges experienced by local businesses that still rely on manual processes for sales and inventory management. The researchers found that manual recording often results in inaccurate inventory data, stock shortages, delayed transactions, and inefficient business operations. The study concluded that implementing an automated inventory and point-of-sale system improves transaction processing, inventory monitoring, and report generation, leading to better business performance.

Aquino, R., & Mendoza, P. (2020). *Inventory and point-of-sale system for Bida Mart. International Journal of Advanced Research in Computer Science*, 11(4), 45-52.

<https://ojs.aaresearchindex.com/index.php/aasgbcjpmra/article/view/13558>

Another study by Barroga, Wabina, and Sales (2021) examined the level of technology adoption among micro, small, and medium enterprises (MSMEs) in the Philippines. The researchers found that many MSMEs still have limited knowledge and awareness regarding advanced technologies such as Business Intelligence systems and web-based platforms. The study identified financial limitations, lack of technical expertise, and resistance to technological change as major barriers affecting digital transformation among local businesses. The researchers emphasized that adopting digital systems can improve productivity, competitiveness, and business sustainability.

Barroga, E., Wabina, M., & Sales, J. (2021). Technology adoption among micro, small, and medium enterprises in the Philippines. *Philippine Journal of Information Technology*, 14(2), 33-47.

<https://ejournals.ph/article.php?id=16789>

ISSN 1908-554X

A related study conducted by Dela Cruz and Santos (2020) focused on the development of a web-based inventory management system for small businesses. The study revealed that implementing a web-based system improved inventory tracking, minimized data redundancy, reduced human

errors, and enhanced operational efficiency. The researchers concluded that web-based systems are practical and cost-effective solutions for small businesses because they allow users to access business data in real time using internet-enabled devices.

Dela Cruz, M., & Santos, R. (2020). Development of a web-based inventory management system for small businesses. *Asia Pacific Journal of Multidisciplinary Research*, 8(3), 112-120.
<https://hal.science/hal-05069591v1/file/Development%20of%20a%20Web%20Based%20Inventory%20Management%20System%20for%20Small%20Businesses.pdf>

Furthermore, a study by Ali et al. (2022) discussed the importance of attendance management systems in improving organizational and institutional performance. The researchers explained that traditional attendance methods are often time-consuming and inefficient, leading to difficulties in monitoring and recording attendance accurately. The study reviewed different automatic identification technologies used in attendance systems and highlighted their potential in managing, recording, and tracking user attendance in various domains. The findings revealed that automated attendance systems provide better efficiency, accuracy, and convenience compared to manual methods, making them beneficial for organizations, industries, and educational institutions.

Ali, N. S., Alhilali, A. H., Rjeib, H. D., & Al-Sadawi, B. (2022). A systematic review of attendance management systems based on identification technologies. *Journal of Physics: Conference Series*, 1228(1), 012038.
https://www.researchgate.net/publication/358178607_Automated_attendance_management_systems_systematic_literature_review
ISSN 1742-6596.

A study conducted by Fernandez and Ramirez (2021) focused on the implementation of a computerized sales and inventory system for small retail businesses in the Philippines. The researchers found that businesses using manual recording methods often experience delays in transaction processing, inaccurate inventory counts, and difficulty in generating reports. The study concluded that computerized systems improve transaction accuracy, inventory monitoring, and report generation, which leads to better operational efficiency and customer service.

Fernandez, P., & Ramirez, L. (2021). Computerized sales and inventory system for small retail businesses. *Philippine Journal of Computing and Information Sciences*, 9(1), 40-51.

<https://ejournals.ph/article.php?id=17221>

ISSN 2244-8156

Another study by Villanueva and Cruz (2020) examined the effectiveness of web-based management systems among local food businesses. The study revealed that many food establishments still rely on manual methods in monitoring sales and inventory, which often results in inaccurate records and delayed transactions. The researchers emphasized that web-based systems improve accessibility, data management, and operational monitoring. The findings also showed that business owners were able to make faster and more accurate decisions through real-time reports generated by the system.

Villanueva, J., & Cruz, M. (2020). Web-based management system for local food businesses in the Philippines. *Asia Pacific Journal of Academic Research in Business Administration*, 6(2), 15-25.

<https://www.apjarba.com/article/view/web-based-management-system>

ISSN 2545-9043

A study conducted by Bautista et al. (2019) explored the implementation of computerized attendance management systems in local enterprises. The researchers found that manual attendance tracking consumes time and often leads to incorrect attendance records and payroll issues. The study concluded that automated attendance systems improve employee monitoring, reduce administrative work, and provide more accurate attendance reports.

Bautista, R., Flores, C., & Hernandez, P. (2019). Computerized attendance management system for local enterprises. *Journal of Information Systems Research and Innovation*, 4(1), 28-36.

<https://www.jisri.org/article/view/attendance-management-system>

ISSN 2467-6748

Related Literature

The adoption of digital technologies in business operations has become increasingly important, particularly for micro, small, and medium enterprises (MSMEs). According to the Philippine Institute for Development Studies (PIDS, 2025), many SMEs are aware of the benefits of digital transformation but face challenges such as high implementation costs, limited technical knowledge, and resistance to change. These barriers prevent businesses from fully utilizing digital tools to improve efficiency and competitiveness.

Inventory management systems are essential for businesses that handle physical products.

According to Stevenson (2018), inventory systems help businesses track stock levels, manage supply chains, and prevent stock shortages. Automated inventory systems provide real-time updates, reducing the risk of overstocking or understocking. Point-of-sale (POS) systems play a critical role in sales management. Laudon and Laudon (2016) explained that POS systems automate the recording of sales transactions, generate receipts, and provide sales reports. These systems improve transaction speed, accuracy, and customer service.

Attendance monitoring systems are also important for managing employees. According to Mathis and Jackson (2011), automated attendance systems improve the accuracy of employee records and reduce the time required for administrative tasks. These systems also help ensure compliance with company policies and labor regulations. Web-based systems have become widely used due to their accessibility and flexibility. According to Shelly and Rosenblatt (2012), web-based applications allow users to access systems through a web browser without the need for installation. This makes them cost-effective and easy to maintain.

Business Intelligence systems integrate data from different sources to provide meaningful insights. Turban et al. (2011) stated that BI systems help organizations analyze data, identify trends, and make informed decisions. These systems are particularly useful for businesses that need to monitor performance and improve operations.

Synthesis of Reviewed Literature

The reviewed literature and studies highlight the growing importance of digital technologies in improving business operations. Foreign studies emphasize the role of business intelligence systems, web-based platforms, and integrated information systems in enhancing efficiency, accuracy, and decision-making. These studies demonstrate that automated systems reduce manual work, minimize errors, and provide real-time access to data.

Local studies, on the other hand, reveal that many MSMEs in the Philippines still rely on manual processes, which lead to inefficiencies and inaccurate records. The studies also highlight challenges such as limited resources, lack of technical expertise, and resistance to change. Despite these challenges, the findings suggest that implementing web-based and integrated systems can significantly improve business performance.

The literature also shows that there is a gap in the availability of affordable and user-friendly digital solutions tailored for small businesses. While many systems exist, they are often not accessible or suitable for SMEs. This gap highlights the need for a system that integrates sales, inventory, and attendance management into a single platform.

Therefore, this study aims to develop a Web-Based Mr. Softy Integrated Business Intelligence Platform that addresses these challenges by providing an efficient, accurate, and user-friendly system for managing business operations. The proposed system is expected to improve data management, reduce manual workload, and support better decision-making for the business.

CHAPTER 3

METHODOLOGY

Research Design

This study uses a developmental research design in creating a Web-Based Mr. Softy Integrated Business Intelligence Platform. This research design is suitable because the study focuses on the design, development, and evaluation of a system that will help improve the business operations of Mr. Softy. The system aims to solve problems related to manual inventory checking, manual sales recording, and manual attendance tracking.

The researchers will first gather information about the current process and identify the problems experienced by the business. After identifying the system requirements, the researchers will design and develop the web-based platform using appropriate programming languages, web technologies, and database management tools. The developed system will include features for inventory management, sales monitoring, attendance tracking, and report generation.

After the development process, the system will be tested and evaluated based on functionality, usability, efficiency, and reliability to determine if it meets the needs of the business. The evaluation results will also be used to improve the system and ensure that it provides a more efficient and accurate way of managing the operations of Mr. Softy.

System Development Methodology

This study will use the Agile methodology in developing the Web-Based Mr. Softy Integrated Business Intelligence Platform. Agile methodology is suitable for this study because it allows continuous development, testing, and improvement of the system based on the needs and feedback of the users. The development process will be divided into several phases to ensure that the system is properly planned, developed, and evaluated.

Requirements Analysis

In this phase, the researchers will gather and analyze the information needed for the system. Interviews, observations, and

consultations with the business owner and users will be conducted to identify the current problems and determine the system requirements. The researchers will focus on issues related to manual sales recording, inventory monitoring, and attendance tracking.

System Design

After gathering the requirements, the researchers will create the design of the system. This includes designing the user interface, database structure, system architecture, and process flow of the platform. Diagrams such as Data Flow Diagrams (DFD), Entity Relationship Diagrams (ERD), and system flowcharts may also be prepared during this phase.

Development

In this phase, the actual coding and development of the system will be performed using appropriate programming languages, frameworks, and database management tools. The features of the platform, such as inventory management, sales tracking, attendance monitoring, and report generation, will be implemented based on the approved system design.

Testing

The developed system will undergo testing to identify errors, bugs, and other technical issues. The researchers will evaluate the functionality, usability, reliability, and efficiency of the system to ensure that all features are working properly and meet the requirements of the business.

Deployment (Prototype)

Once the system passes the testing phase, the prototype version of the platform will be deployed for actual use and demonstration. The users will be allowed to interact with the system and provide feedback regarding its performance and usability in real business operations.

Continuous Improvement

Feedback and evaluation results gathered from users during the deployment phase will be used to improve the system. Necessary

modifications, updates, and enhancements will be applied to further increase the efficiency, usability, and overall performance of the platform.

System Architecture

The Mr. Softy Web-Based Integrated Business Intelligence Platform follows a **client-server architecture**. In this type of architecture, the client side allows users to access and interact with the system through a web browser, while the server side is responsible for processing requests, handling business logic, managing the database, and generating reports and analytics.

The client sends requests such as recording sales, checking inventory, or monitoring attendance to the server. The server then processes the data and stores it in the database before sending the results back to the user interface. This architecture helps the system provide faster data processing, centralized data management, and secure access for multiple users at the same time.

The client-server architecture is suitable for the study because it supports real-time data access, easier system maintenance, and improved scalability. It also allows the researchers to manage and update the system efficiently without affecting the users' access to the platform.

Tool and Technology Used

- **Frontend:** HTML, CSS, JavaScript
- **Backend:** PHP, MySQL
- **Database:** MySQL Work Bench
- **Development Tools:** Visual Studio Code

Data Gathering Techniques

- **Interviews**
- **Observation**

Testing Procedures

- **Unit Testing-** Test individual parts of the system.
- **Integration Testing-** Test how different parts work together.
- **System Testing-** Test the whole system as one.
- **User Acceptance Testing-** Let the user/client test the system.

Evaluation Criteria

- **Functionality-** Features work correctly and perform intended tasks.
- **Usability-** The system is easy to use and navigate.
- **Reliability-** The system operates consistently without crashing.
- **Security-** Data is protected and inputs are validated.
- **Efficiency-** The system responds quickly and uses resources effectively.

SUPPORTING DIAGRAMS

DATA FLOW DIAGRAM

User → Mr. Softy System → Database

- **Staff** interacts with the system to input customer orders and payments.
- **The system** handles order processing, inventory updates, and receipt generation.
- **Database** stores all order, payment, and product information.
- **Admin** manages inventory and accesses reports; analytics are generated and stored for decision-making.

ENTITY RELATIONSHIP DIAGRAM

Entities:

- **User**(UserID, Name, Username, Password, StoreID, Role)
- **Store**(StoreID, StoreName, Location)
- **Inventory**(InventoryID, StoreID, ItemID, Quantity, LowStockLevel, UpdatedAt)
- **Products**(ProductID, ProductName, Size, Price, Category, Status, CreatedAt)
- **ToppingInventory**(ToppingInventoryID, BranchID, ToppingID, Quantity, LowStockLevel, UpdatedAt)
- **StockRequest**(RequestID, StoreID, ItemID, RequestedQty, Status, RequestedBy, CreatedAt)
- **Item**(ItemID, ItemName, Category)
- **ProductItems**(ProductItemID, ProductId, ItemID, QuantityNeeded)
- **Notification**(NotifID, UsedID, StoreID, RequestID, ItemName, RequestedQty, Status, IsRead, CreatedAt, UpdatedAt)
- **StockCheckLogs**(LogID, StoreID, StaffID, ItemID, CheckedAt, Remarks, CheckDate)
- **Attendance**(AttendanceID, UserID, StoreID, Role, Date, TimeIn, TimeOut)
- **Toppings**(ToppingID, ToppingName, Price, Status, CreatedAt)
- **Sales**(SaleID, ProductID, CashierID, StoreID, Quantity, Subtotal, SaleDate)
- **SaleToppings**(SaleToppingID, SaleID, ToppingID)
- **Orders**(OrderID, CashierID, StoreID, TotalAmount, OrderDate, OrderTime, CreatedAt)

Relationships:

- A **User** works at a **Store**.
- A **Store** has **Items** in its **Inventory** and **Toppings** in **ToppingInventory**.
- A **Product** is made from **Items** (through **ProductItems**).
- A **User** can request more **Items** via a **StockRequest**, which can trigger a **Notification**.
- **Users** can log **StockChecks** and **Attendance** at a **Store**.
- A **Sale** happens when a **User** sells a **Product** at a **Store**.
- **SaleToppings** link **Sales** to the **Toppings** sold with a **Product**.
- An **Order** contains multiple **Sales** and happens at a **Store**.

USE CASE DIAGRAM

Actors:

- **Staff / User**
- **Admin**

Use Cases:

Staff / User

- Log In to the System
- Add Orders for Customers (Products + Toppings)
- View and Manage Orders
- Void Incorrect Items
- Process Checkout and Billing
- Check Inventory
- Create Stock Requests
- Log Stock Checks
- Record Attendance

Administrator:

- Log In to the System
- Manage Products and Toppings
- View Sales Reports
- Monitor Transactions and Orders
- Manage User Accounts

- Oversee Store Operations and Notifications